SAANVI KANODIA

Data Science & Machine Learning | Applied AI & Research

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EDUCATION

MPSTME, NMIMS *July 2023 – Ongoing*

B.Tech in Computer Science Engineering (Data Science) — CGPA: 3.76/4 Mumbai, India

INTERNSHIP EXPERIENCE

Data Engineering Intern — PwC *May 2025 – August 2025*

- Developed a multi-agent GenAI pipeline using Azure OpenAI and AutoGen to transform unstructured PDFs into structured databases through OCR, JSON conversion, summarization, and SQLite ingestion, reducing manual effort by 70%.
- Benchmarked 5+ OCR models (GPT-4o, PaddleOCR, EasyOCR, Pytesseract) to select the optimal extraction strategy, improving text extraction accuracy by **25%**.

Project Management Intern — Concept Simplified *May 2024 – Sep 2024*

- Led a team of 10 interns for MHT-CET counselling, guiding 600+ students through admissions decisions.
- Created engineering-focused content for the company's YouTube page, helping grow the subscriber base to 50k.

PROJECTS

Computer Vision for Building Defect Classification – *(Python, CNNs, Image Processing)* GitHub

- Developed a hybrid defect-classification pipeline combining CNN-based image learning with HSV/CLAHE preprocessing and classical image features for **damp, mold, crack, spalling, and peeling** detection.
- Compared Random Forest and XGBoost against the CNN pipeline, improving validation F1 from **0.736 → 0.755** with feature-based modeling.

Multimodal Document Intelligence – *(Python · PyMuPDF · Pytesseract · OpenCV · LLaMA)* GitHub

- Built a multimodal PDF-processing pipeline using PyMuPDF and Pytesseract to extract text and images, with OpenCV/Pillow-based image processing for OCR.
- Integrated a local TinyLlama 1.1B model via to perform contextual understanding and summarisation of extracted document content.

Agentic Fraud Detection System – *(Python, LangGraph, scikit-learn, PySpark)* GitHub

- Built a modular **LangGraph multi-agent pipeline** spanning data ingestion, EDA, model training, prediction, and evaluation for financial fraud detection.
- Implemented the pipeline with **PySpark MLlib**, reducing false negatives **by 15% compared with the scikit-learn implementation**

Agentic AI Support System for E-Commerce – (*Python, LLM Agents)* GitHub

- Built a hybrid ML and multi-agent system for e-commerce migration incidents, combining clustering, classification, semantic search, and RAG-based resolution.
- Designed Investigator, Researcher, Analyst, and Responder agents with confidence-based routing, human review, automated email responses, and GitHub-based documentation updates.

SKILLS

Languages: Python, SQL

ML / AI: scikit-learn, PySpark MLlib, LangChain, LangGraph, AutoGen, Azure OpenAI, Computer Vision (BLIP, CNNs), LLMs

Data / BI : Databricks, Apache Airflow, Tableau, Power BI

ACHIEVEMENTS & CERTIFICATIONS

- **Chairperson** — Association of Computing Machinery (ACM)
- Supervised Machine Learning: Regression and Classification — Coursera (Stanford & DeepLearning.AI)